TUGAS KONSEP JARINGAN

TUGAS KE – 3 Classless IP



Nama : Shevanka Bagus Dwiyudi Kurniawan

NRP : 3125600044

Dosen Pengajar : Dr. Ferry Astika Saputra, ST, M.Sc

PROGRAM STUDI D4 TEKNIK INFORMATIKA

POLITEKNIK ELEKTRONIKA NEGERI SURABAYA (PENS)

TAHUN 2026

A. Tugas

1. Gambar topologi 192.168.1.0/24 dengan 4 segmentasi jaringan
2. Topologi jaringan dengan IP 132.10.0.0/16 dengan 10 segmentasi jaringan
3. Topologi jaringan 17.8.0.0/16 dengan 4 segmentasi jaringan
4. Topologi jaringan 8.32.0.0 dengan 6 segmentasi jaringan

B. Jawaban

1. 192.168.1.0/24 dengan 4 segmentasi

   a. Perhitungan Subnetting

   - Alamat Dasar: 192.168.1.0/24 (Kelas C, default mask 255.255.255.0).
   - Kebutuhan Subnet: 4 segmentasi jaringan valid.
   - Perhitungan Bit Subnet (n):

     Meminjam $n = 3$ bit:

     Σ subnet valid $= 2^3 - 2 = 6$ subnet valid (mencukupi 4 segmen)
   - Prefix Baru: $/24 + 3 = /\mathbf{27}$.
   - Subnet Mask Baru: 255.255.255.224

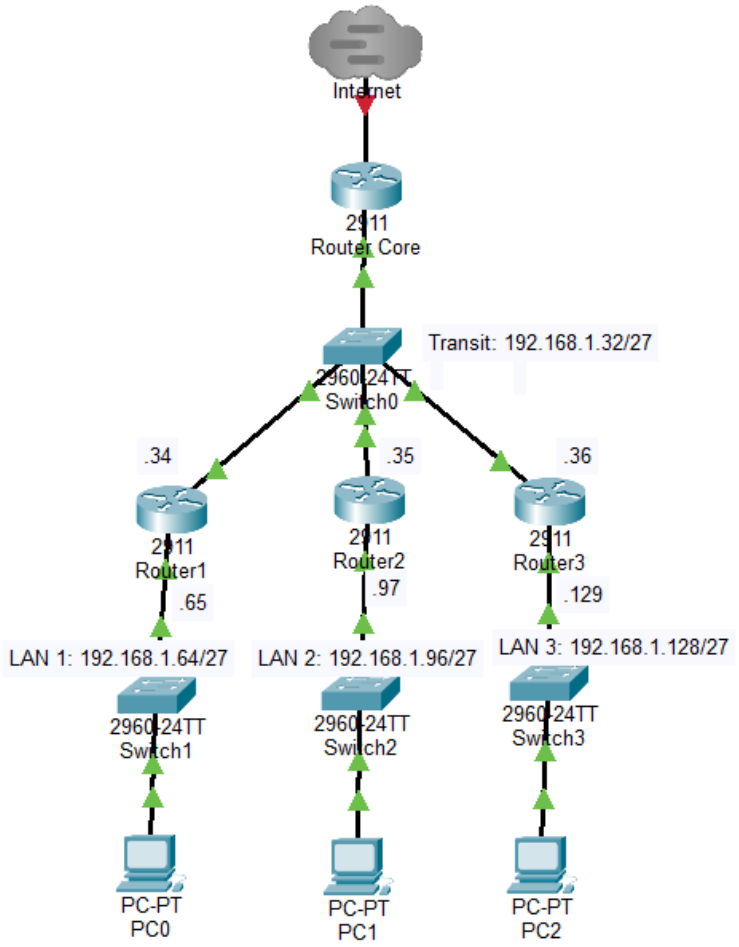
     (biner: 11111111.11111111.11111111.11100000)
   - Interval Blok Subnet (Step): $256 - 224 = 32$.
   - Jumlah Host per Subnet: $2^5 - 2 = 30$ host usable.

   b. Tabel Alokasi 4 Subnet

No	Nilai Biner	Network Address	Host Pertama (Gateway)	Host Terakhir	Broadcast Address
-	000	192.168.1.0/27	192.168.1.1	192.168.1.30	192.168.1.31
1	001	192.168.1.32/27	192.168.1.33	192.168.1.62	192.168.1.63
2	010	192.168.1.64/27	192.168.1.65	192.168.1.94	192.168.1.95
3	011	192.168.1.96/27	192.168.1.97	192.168.1.126	192.168.1.127

4	100	192.168.1.128/27	192.168.1.129	192.168.1.158	192.168.1.159
-	101	192.168.1.160/27	192.168.1.161	192.168.1.190	192.168.1.191
-	110	192.168.1.192/27	192.168.1.193	192.168.1.222	192.168.1.223
-	111	192.168.1.224/27	192.168.1.225	192.168.1.254	192.168.1.255

c. Skema Diagram Topologi



2. 132.10.0.0/16 dengan 10 segmentasi

a. Perhitungan Subnetting

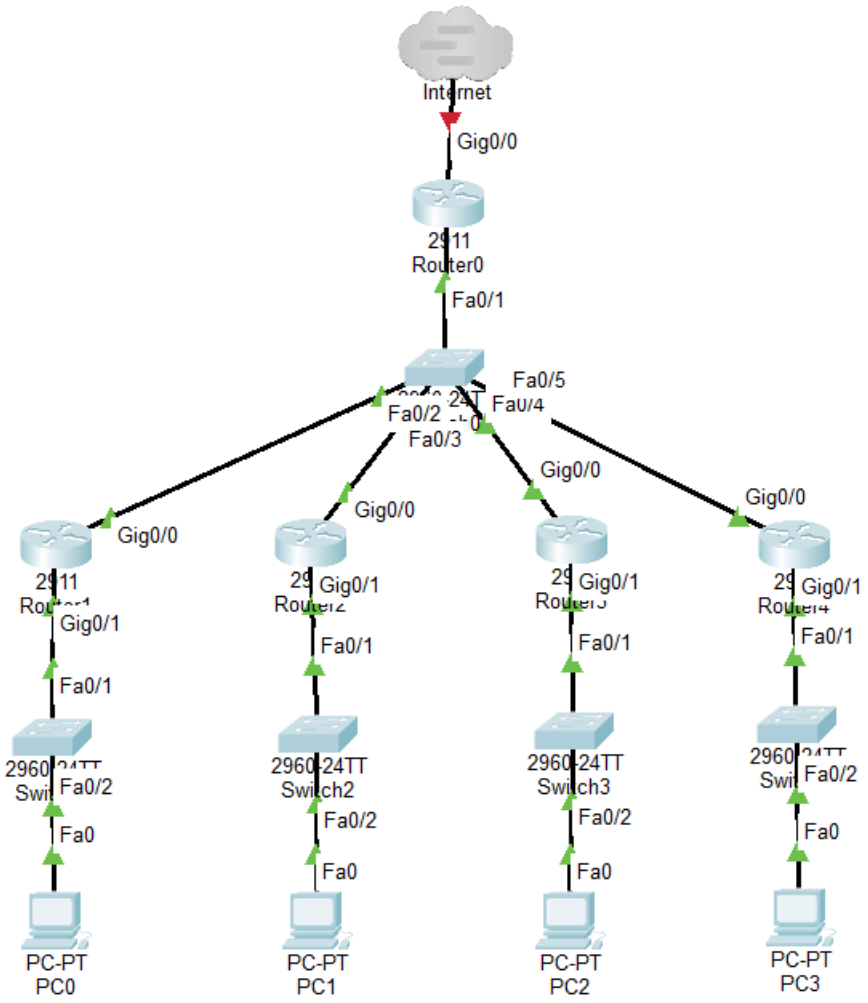
- Alamat Dasar: 132.10.0.0/16 (Kelas B, default mask 255.255.0.0).
- Kebutuhan Subnet: 10 segmentasi jaringan.
- Jumlah Bit yang Dipinjam (n):

  $2^n \geq 10 \Longrightarrow n = 4$ $(2^4 = 16$ subnet) Prefix Baru: $/16 + 4 = /20$.
- Subnet Mask Baru: 255.255.240.0 (biner oktet ke-3: 11110000).
- Interval Blok Subnet (Step): $256 - 240 = 16$ pada oktet ke-3.
- Jumlah Host per Subnet: $2^{12} - 2 = 4.094$ host usable.

b. Tabel Alokasi 6 Subnet

No	Biner	Network Address	Host Pertama (Gateway)	Host Terakhir	Broadcast Address
-	0000	132.10.0.0/20	132.10.0.1	132.10.15.254	132.10.15.255
1	0001	132.10.16.0/20	132.10.16.1	132.10.31.254	132.10.31.255
2	0010	132.10.32.0/20	132.10.32.1	132.10.47.254	132.10.47.255
3	0011	132.10.48.0/20	132.10.48.1	132.10.63.254	132.10.63.255
4	0100	132.10.64.0/20	132.10.64.1	132.10.79.254	132.10.79.255
5	0101	132.10.80.0/20	132.10.80.1	132.10.95.254	132.10.95.255
6	0110	132.10.96.0/20	132.10.96.1	132.10.111.254	132.10.111.255
7	0111	132.10.112.0/20	132.10.112.1	132.10.127.254	132.10.127.255
8	1000	132.10.128.0/20	132.10.128.1	132.10.143.254	132.10.143.255
9	1001	132.10.144.0/20	132.10.144.1	132.10.159.254	132.10.159.255
10	1010	132.10.160.0/20	132.10.160.1	132.10.175.254	132.10.175.255
-	...	Cadangan (1011–1110)	...	...	...
-	1111	132.10.240.0/20	132.10.240.1	132.10.255.254	132.10.255.255

c. Skema Diagram Topologi



3. 17.8.0.0/16 dengan 4 segmentasi

a. Perhitungan Subnetting

- Alamat Dasar: 17.8.0.0/16 (Kelas B, default mask /16).
- Kebutuhan Subnet: 4 segmentasi valid.
- Jumlah Bit Subnet (n):

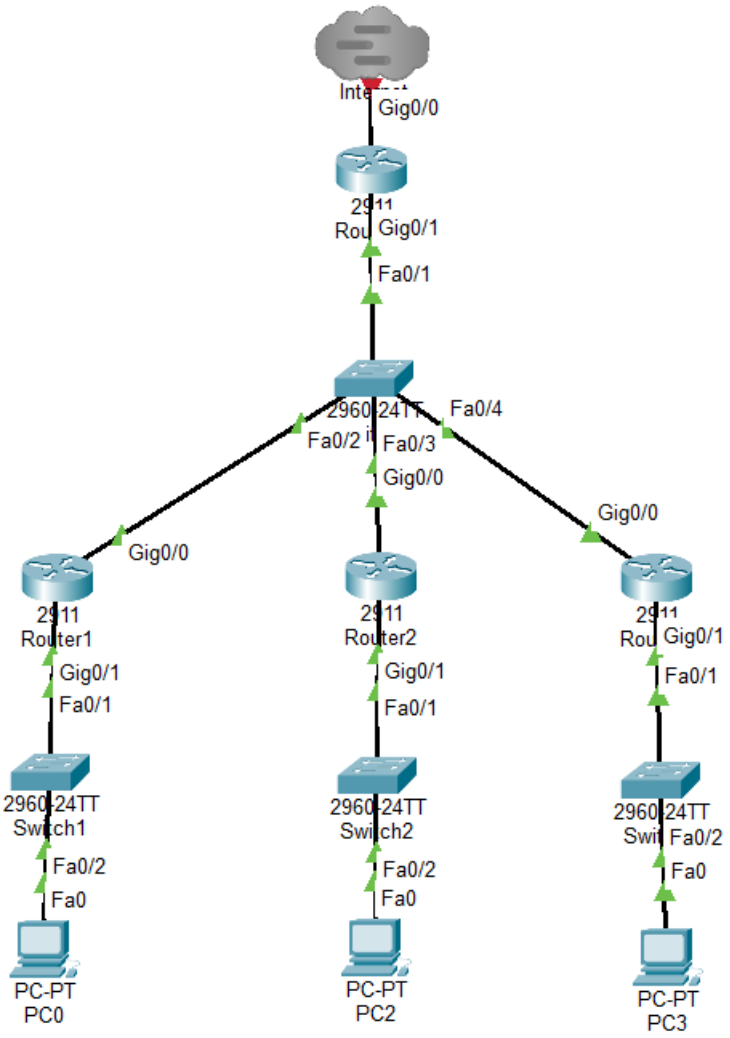
Maka meminjam $n = 3$ bit: $2^3 - 2 = 6$ subnet valid (mencukupi 4 segmen).

- Prefix Baru: $/16 + 3 = /\mathbf{19}$.
- Subnet Mask Baru: 255.255.224.0 (biner oktet ke-3: 11100000).
- Interval Blok Subnet (Step): $256 - 224 = 32$ pada oktet ke-3.
- Jumlah Host per Subnet: $2^{13} - 2 = 8.190$ host usable.

b. Tabel Alokasi 6 Subnet

No	Biner	Network Address	Host Pertama (Gateway)	Host Terakhir	Broadcast Address
-	000	17.8.0.0/19	17.8.0.1	17.8.31.254	17.8.31.255
1	001	17.8.32.0/19	17.8.32.1	17.8.63.254	17.8.63.255
2	010	17.8.64.0/19	17.8.64.1	17.8.95.254	17.8.95.255
3	011	17.8.96.0/19	17.8.96.1	17.8.127.254	17.8.127.255
4	100	17.8.128.0/19	17.8.128.1	17.8.159.254	17.8.159.255
-	...	Cadangan (101–110)	...	...	...
-	111	17.8.224.0/19	17.8.224.1	17.8.255.254	17.8.255.255

c. Skema Diagram Topologi



4. 8.32.0.0 dengan 6 segmentasi

a. Perhitungan Subnetting

- Alamat Dasar: 8.32.0.0/12
- Kebutuhan Subnet: 6 segmentasi valid.
- Jumlah Bit Subnet (n):

  $2^n - 2 \geq 6 \Rightarrow 2^n \geq 8 \Rightarrow n = 3$ bit
- Prefix Baru: $/12 + 3 = /\mathbf{15}$.

- Subnet Mask Baru:
  - Oktet 1: 11111111 = 255
  - Oktet 2: 4 bit asal (1111) + 3 bit subnet (111) + 1 bit host (0) = 11111110 = 254
  - Subnet Mask: 255.254.0.0.
- Interval Blok Subnet (Step): $256 - 254 = 2$ pada oktet ke-2.
- Jumlah Host per Subnet: $2^{17} - 2 = 131.070$ host usable.

b. Tabel Alokasi 6 Subnet

No	Biner	Network Address	Host Pertama (Gateway)	Host Terakhir	Broadcast Address
-	000	8.32.0.0/15	8.32.0.1	8.33.255.254	8.33.255.255
1	001	8.34.0.0/15	8.34.0.1	8.35.255.254	8.35.255.255
2	010	8.36.0.0/15	8.36.0.1	8.37.255.254	8.37.255.255
3	011	8.38.0.0/15	8.38.0.1	8.39.255.254	8.39.255.255
4	100	8.40.0.0/15	8.40.0.1	8.41.255.254	8.41.255.255
5	101	8.42.0.0/15	8.42.0.1	8.43.255.254	8.43.255.255
6	110	8.44.0.0/15	8.44.0.1	8.45.255.254	8.45.255.255
-	111	8.46.0.0/15	8.46.0.1	8.47.255.254	8.47.255.255

c. Skema Diagram Topologi

